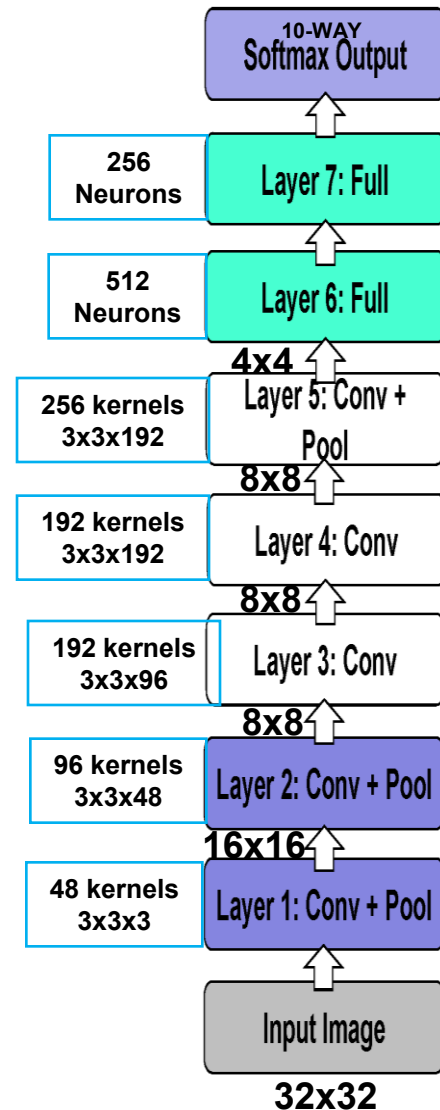


HW# 5

Deep Learning for Computer Vision (EN525.733-8VL)

Due date: March 12th, 2026

1. Use CIFAR-10 data set of resolution (32x32) and train an AlexNet-like network as shown in the diagram on the right for 10 classes. **Plot learning curves and classification performance + confusion matrix.** Use the training and testing dataset specified in the CIFAR-10 database (<https://www.cs.toronto.edu/~kriz/cifar.html>)
 - Use the kernel size 3x3 and the number of kernel as shown in the diagram.
 - Replace the FC layers of AlexNet to FC-512, FC-256 as shown in the diagram.
2. Repeat the experiment (retrain) with deleting only the Layer 5 Conv_256 + its max-pooling. **Plot learning curves and classification performance + confusion Matrix.**
3. Repeat the experiment with deleting only the layer 4 Conv-192. **Plot learning curves and classification performance + confusion matrix**
4. Repeat the experiment with deleting the FC layer 6. **Plot learning curves and classification performance + confusion matrix.**



<https://www.cs.toronto.edu/~kriz/cifar.html>

http://rodrigob.github.io/are_we_there_yet/build/classification_datasets_results.html